Intelligent Credit Risk Analysis for Accurate and Explainable Loan Applicant Classification

Final Project Review | Department of Computer Science and Engineering | Koneru Lakshmaiah Education Foundation

TEAM MEMBERS

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Guide: Ms. Prathusha, Assistant Professor | Academic Year: 2026–2027

Introduction & Problem Statement



Background

Credit-risk assessment identifies applicants who may struggle to repay. ML improves classification, but accuracy alone can hide the unequal cost of different errors.



Problem Statement

Conventional models focus on accuracy without weighing asymmetric misclassification cost or explaining why an applicant is high or low risk.



Motivation / Need

Predicting a bad-risk applicant as good costs far more than the reverse. FinSight combines cost-aware evaluation, explainability and human-centered decision support.

5 : 1

Assumed cost ratio

Bad-risk applicant predicted good (FN) vs. good-risk predicted bad (FP)

Cost = 5·FN + FP

Objectives & Scope

Objectives

1. Develop a leakage-safe preprocessing and credit-risk classification pipeline.
2. Compare Logistic Regression, Decision Tree, Random Forest and Gradient Boosting.
3. Evaluate with Accuracy, Precision, Recall, F1, ROC-AUC and asymmetric misclassification cost.
4. Provide interpretable predictions using SHAP and an interactive Flask decision-support system.

In Scope

Data validation, preprocessing, EDA, model comparison, threshold selection, SHAP, risk scoring, Flask analytics, What-If analysis and AWS S3 / Glue / Athena integration.

Out of Scope

Automated real-world loan approval and production banking deployment.

Literature Survey

S.No	Author(s) & Year	Method / Technique	Key Findings	Limitation / Gap
1	Lundberg & Lee, 2017	SHAP / additive feature attribution	Consistent feature-level explanations	General explanation framework; not credit-cost threshold analysis
2	Bussmann et al., 2020	Explainable AI for fintech risk	Shapley-based explanations improve transparency	No asymmetric threshold-selection experiment
3	Chen, Calabrese & Martin-Barragan, 2024	Interpretable ML for imbalanced credit scoring	Studies interpretation and class imbalance	Cost-aware model selection needs further evaluation
4	Ballegeer, Bogaert & Benoit, 2025	Instance-dependent cost-sensitive scoring; explanation stability	Cost-sensitive learning and explanation reliability matter	Motivates joint evaluation of cost, thresholds and explainability



Research Gap

Prior work treats prediction, explainability, class imbalance and cost-sensitivity separately. FinSight evaluates their interaction through cost-aware threshold selection, model comparison and applicant-level explanation.

Proposed System & Methodology

STEP 1

Data Validation

Validate UCI German Credit schema, labels, missing values and duplicates.

STEP 2

Preprocessing

Encode categoricals, process numerics, build stratified train/test sets.

STEP 3

Model Development

Train Logistic Regression, Decision Tree, Random Forest and Gradient Boosting.

STEP 4

Cost-Aware Evaluation

Compare metrics and asymmetric cost; select thresholds from training evidence.

STEP 5

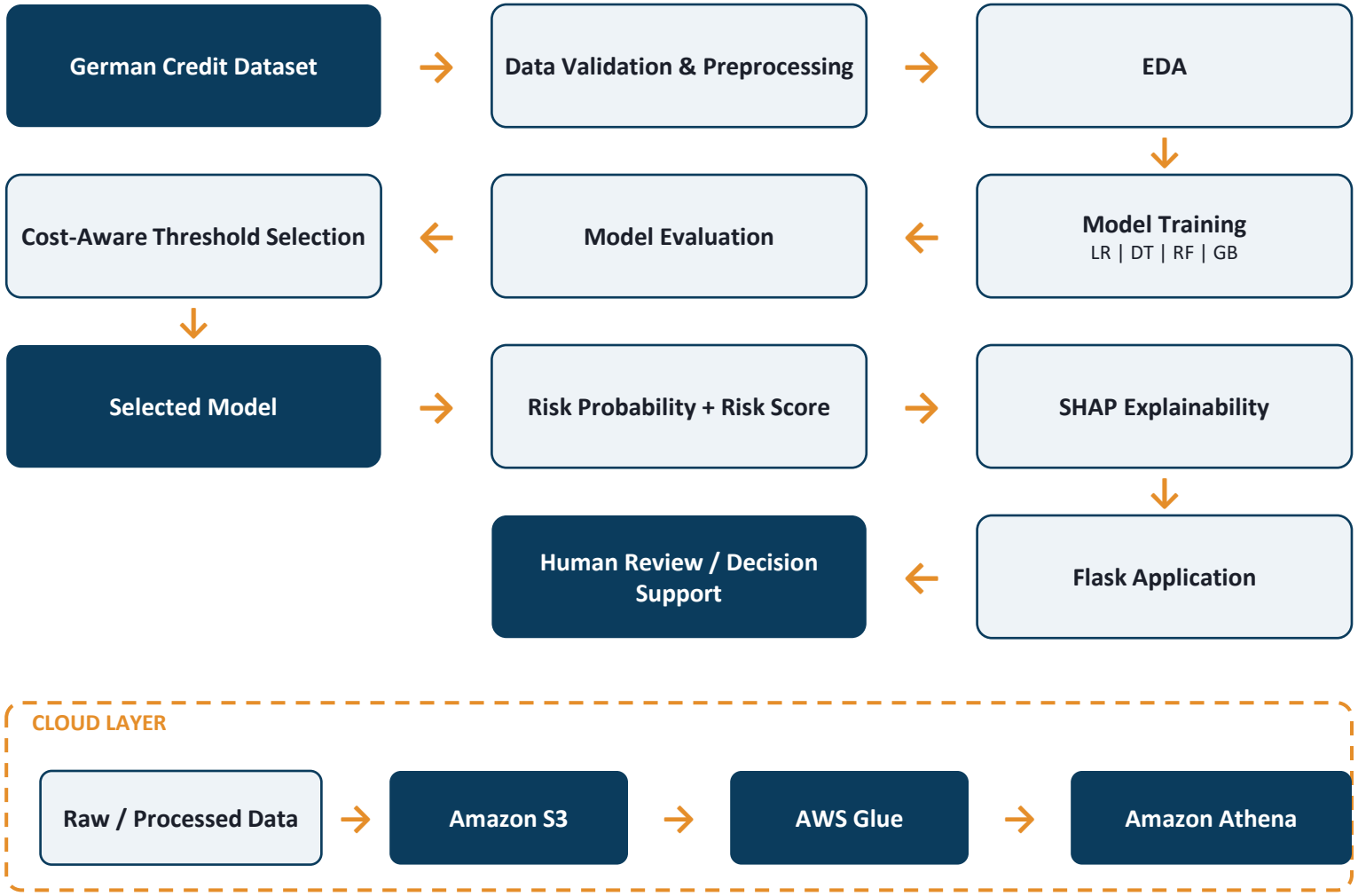
Explain & Deploy

Generate SHAP explanations and expose results via the Flask dashboard.

Key Novelty / Contribution

FinSight evaluates credit-risk models on predictive performance and asymmetric error cost together, not accuracy alone, and adds model explanations for human review.

System Architecture



Components

1. **Data Layer**
   Validation, preprocessing, analytics data

2. **ML Layer**
   Training, evaluation, threshold selection

3. **Explainability Layer**
   SHAP, probability and risk score

4. **Application Layer**
   Flask dashboard and cloud analytics

Hardware & Software Requirements

Hardware

- Standard laptop / desktop
- 8 GB RAM recommended
- Multi-core CPU
- Internet connection
- No dedicated GPU required

Software & Tools

- Windows
- Python
- VS Code
- Git / GitHub
- Web browser

Libraries / Frameworks

- Pandas, NumPy, Scikit-learn
- SHAP, Joblib, Pytest
- Flask, Plotly, Bootstrap
- Boto3
- Cloud: Amazon S3, AWS Glue, Athena

Platform: Local Flask application with AWS cloud analytics support.

Dataset / Data Understanding

1,000

Records (rows)

20

Predictor features

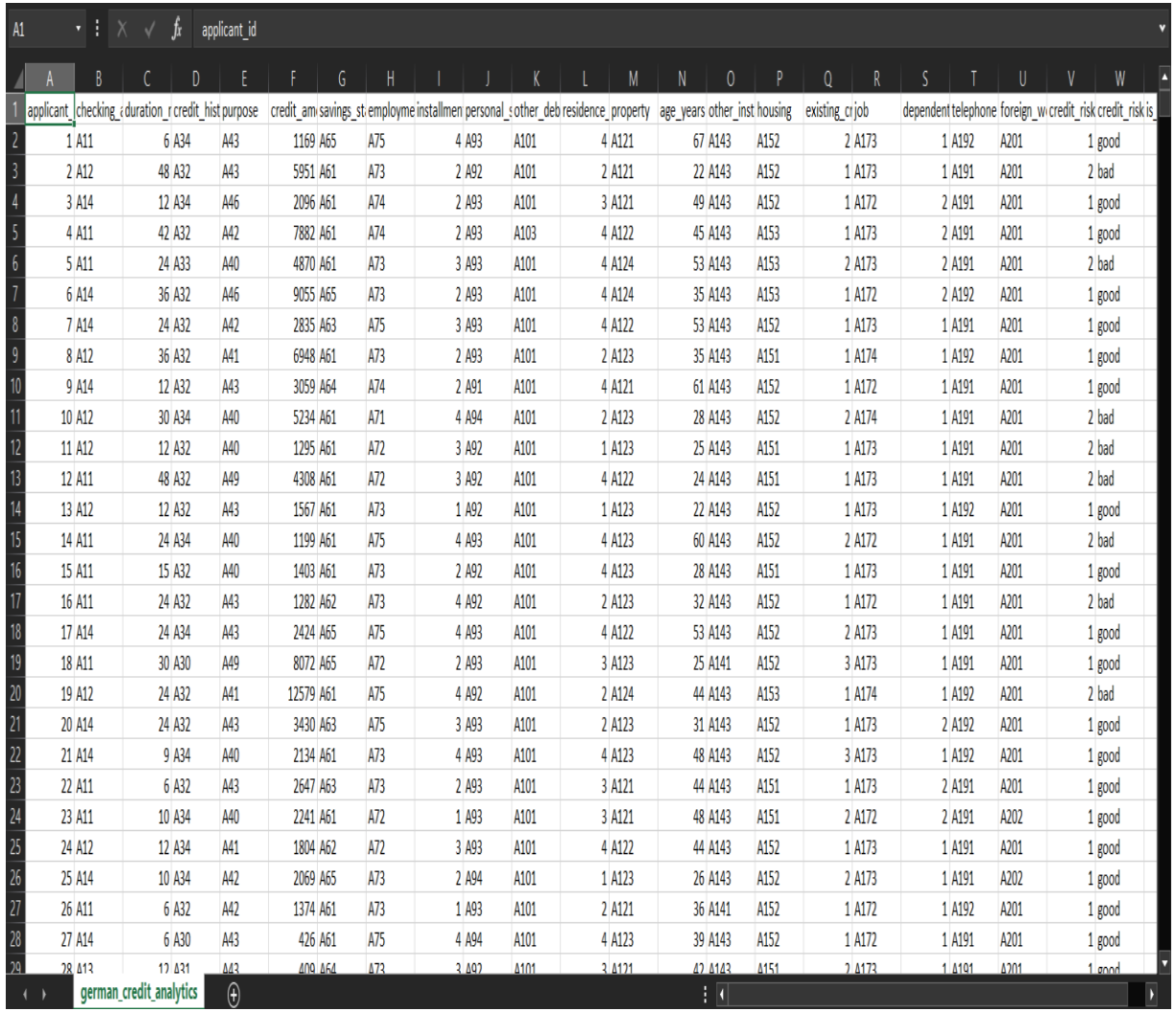
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Missing values

2

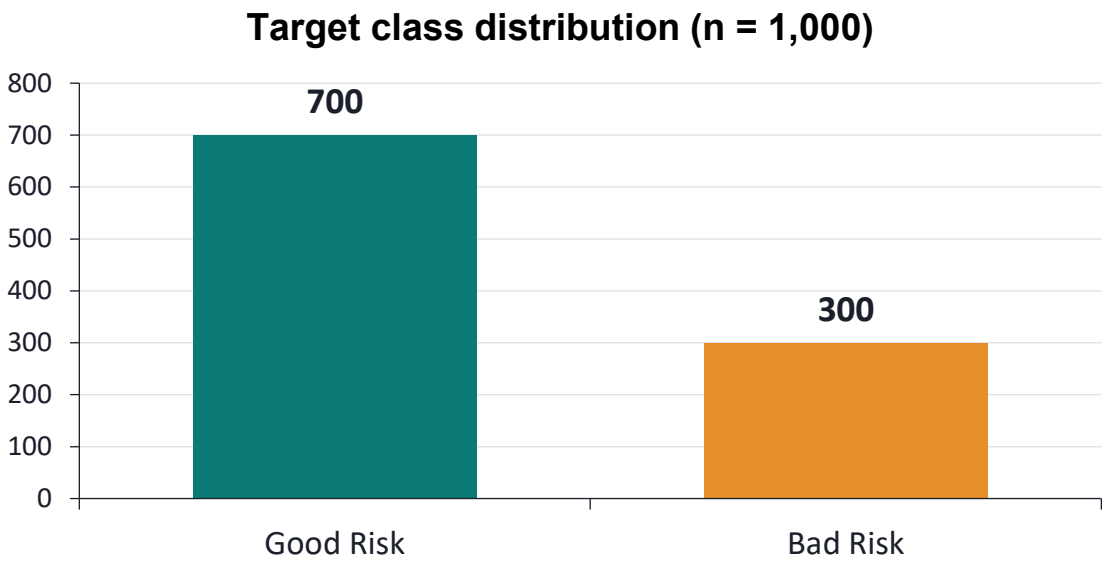
Target classes

- Source: UCI German Credit Dataset
- Classes: 700 Good Risk / 300 Bad Risk
- Features: checking status, duration, credit history, purpose, credit amount, savings, employment, age
- Quality: schema validated, no missing values, 70:30 imbalance considered in evaluation



applicant_	checking_	duration_r	credit_hist	purpose	credit_am	savings_st	employme	installmen	personal_s	other_deb	residence_	property	age_years	other_inst	housing	existing_cr	job	dependent	telephone	foreign_w	credit_risk	credit_risk is_
1	A11	6	A34	A43	1169	A65	A75	4	A93	A101	4	A121	67	A143	A152	2	A173	1	A192	A201	1	good
2	A12	48	A32	A43	5951	A61	A73	2	A92	A101	2	A121	22	A143	A152	1	A173	1	A191	A201	2	bad
3	A14	12	A34	A46	2096	A61	A74	2	A93	A101	3	A121	49	A143	A152	1	A172	2	A191	A201	1	good
4	A11	42	A32	A42	7882	A61	A74	2	A93	A103	4	A122	45	A143	A153	1	A173	2	A191	A201	1	good
5	A11	24	A33	A40	4870	A61	A73	3	A93	A101	4	A124	53	A143	A153	2	A173	2	A191	A201	2	bad
6	A14	36	A32	A46	9055	A65	A73	2	A93	A101	4	A124	35	A143	A153	1	A172	2	A192	A201	1	good
7	A14	24	A32	A42	2835	A63	A75	3	A93	A101	4	A122	53	A143	A152	1	A173	1	A191	A201	1	good
8	A12	36	A32	A41	6948	A61	A73	2	A93	A101	2	A123	35	A143	A151	1	A174	1	A192	A201	1	good
9	A14	12	A32	A43	3059	A64	A74	2	A91	A101	4	A121	61	A143	A152	1	A172	1	A191	A201	1	good
10	A12	30	A34	A40	5234	A61	A71	4	A94	A101	2	A123	28	A143	A152	2	A174	1	A191	A201	2	bad
11	A12	12	A32	A40	1295	A61	A72	3	A92	A101	1	A123	25	A143	A151	1	A173	1	A191	A201	2	bad
12	A11	48	A32	A49	4308	A61	A72	3	A92	A101	4	A122	24	A143	A151	1	A173	1	A191	A201	2	bad
13	A12	12	A32	A43	1567	A61	A73	1	A92	A101	1	A123	22	A143	A152	1	A173	1	A192	A201	1	good
14	A11	24	A34	A40	1199	A61	A75	4	A93	A101	4	A123	60	A143	A152	2	A172	1	A191	A201	2	bad
15	A11	15	A32	A40	1403	A61	A73	2	A92	A101	4	A123	28	A143	A151	1	A173	1	A191	A201	1	good
16	A11	24	A32	A43	1282	A62	A73	4	A92	A101	2	A123	32	A143	A152	1	A172	1	A191	A201	2	bad
17	A14	24	A34	A43	2424	A65	A75	4	A93	A101	4	A122	53	A143	A152	2	A173	1	A191	A201	1	good
18	A11	30	A30	A49	8072	A65	A72	2	A93	A101	3	A123	25	A141	A152	3	A173	1	A191	A201	1	good
19	A12	24	A32	A41	12579	A61	A75	4	A92	A101	2	A124	44	A143	A153	1	A174	1	A192	A201	2	bad
20	A14	24	A32	A43	3430	A63	A75	3	A93	A101	2	A123	31	A143	A152	1	A173	2	A192	A201	1	good
21	A14	9	A34	A40	2134	A61	A73	4	A93	A101	4	A123	48	A143	A152	3	A173	1	A192	A201	1	good
22	A11	6	A32	A43	2647	A63	A73	2	A93	A101	3	A121	44	A143	A151	1	A173	2	A191	A201	1	good
23	A11	10	A34	A40	2241	A61	A72	1	A93	A101	3	A121	48	A143	A151	2	A172	2	A191	A202	1	good
24	A12	12	A34	A41	1804	A62	A72	3	A93	A101	4	A122	44	A143	A152	1	A173	1	A191	A201	1	good
25	A14	10	A34	A42	2069	A65	A73	2	A94	A101	1	A123	26	A143	A152	2	A173	1	A191	A202	1	good
26	A11	6	A32	A42	1374	A61	A73	1	A93	A101	2	A121	36	A141	A152	1	A172	1	A192	A201	1	good
27	A14	6	A30	A43	426	A61	A75	4	A94	A101	4	A123	39	A143	A152	1	A172	1	A191	A201	1	good
28	A13	12	A31	A43	409	A64	A73	3	A92	A101	3	A121	42	A143	A151	2	A173	1	A191	A201	1	good

Data Preprocessing & Exploratory Analysis



EDA Summary

Target distribution: 70:30 (Good : Bad)
Duration: ≈ 4–72 months
Credit amount: ≈ 250–18,424 benchmark units
Age: ≈ 19–75 years

Cleaning

Schema, duplicate, target and numeric-range validation

Transformation

One-hot encoding, imputation where needed, scaling for LR; same fitted transforms reused at prediction

Data Split

80% train / 20% test, stratified, random seed 42

Module-wise Description

1 Core ML & Cost-Aware Evaluation

Developed by: M. Manasvini

Purpose: Core ML training, comparison and final model selection

Responsibilities: ML pipeline; LR, DT, RF, GB; hyperparameter tuning; model comparison; asymmetric cost; threshold selection; final integration

Input → Output: Processed applicant data → trained/evaluated credit-risk model

2 Data Processing, EDA & Explainability

Developed by: N. Sreeja

Purpose: Prepare validated data and interpret model behavior

Responsibilities: Dataset validation; preprocessing; EDA; feature transformation; SHAP; important-feature interpretation

Input → Output: Raw credit data → processed data + explainability outputs

3 Application & Interactive Risk Analysis

Developed by: S. Sreeja

Purpose: Build the user-facing decision-support system

Responsibilities: Flask dashboard; applicant prediction; probability; 0–100 risk score; risk bands; What-If Simulator; Plotly/Bootstrap UI

Input → Output: Applicant input → interactive prediction and risk analysis

4 Cloud Integration, Testing & Documentation

Developed by: M. Sahasra

Purpose: Support cloud analytics, validation and verification

Responsibilities: Amazon S3; AWS Glue; Athena; Pytest; Flask route validation; integration checks; documentation

Input → Output: Project artifacts/data → cloud analytics + validated system

Algorithms & Technologies Used

Logistic Regression

Why chosen
Interpretable linear baseline

Key parameters
C = 0.1

Random Forest

Why chosen
Nonlinear ensemble that models complex interactions

Key parameters
200 trees, max depth 5, balanced class weights

Gradient Boosting

Why chosen
Strong nonlinear ensemble; selected from training-cost evidence

Key parameters
200 estimators, learning rate 0.05, depth 3, subsample 0.8

Selection criteria: Asymmetric training cost, supported by ROC-AUC, recall and F1-score. Decision Tree is an additional baseline.

Implementation / Model Building

Feature Selection

20 raw predictors, one-hot encoded categoricals

Preprocessing Pipeline

Validation → preprocessing → stratified split → training → 5-fold CV → probabilities → cost-aware threshold

Data Splitting

800 training / 200 holdout (80:20, stratified, seed 42)

Configuration

Models: LR, DT, RF, GB; cost = 5·FN + FP

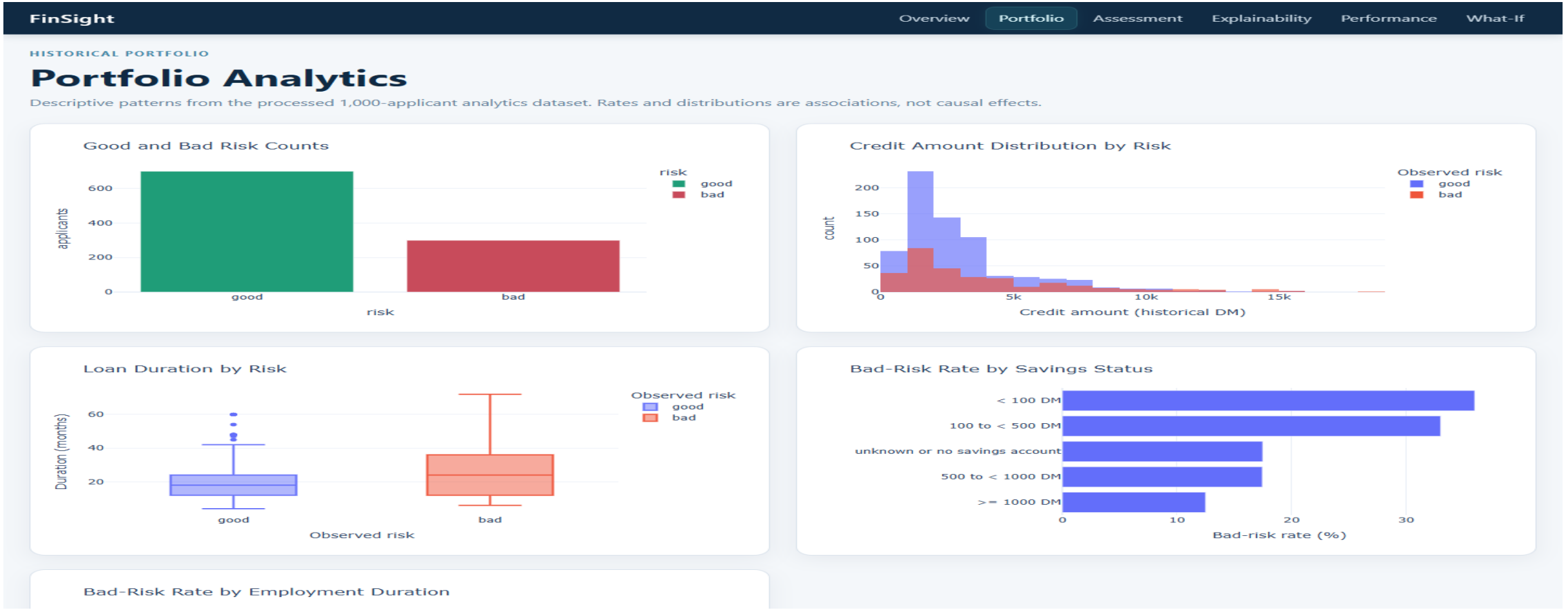
Code snippet / pseudocode

```
# FinSight pipeline (simplified)
X_tr, X_te, y_tr, y_te = train_test_split(
    X, y, test_size=0.2,
    stratify=y, random_state=42)

pipe = Pipeline([
    ("prep",  preprocessor),  # fit on train only
    ("model", model)])         # LR/DT/RF/GB
pipe.fit(X_tr, y_tr)

p_bad = pipe.predict_proba(X_te)[:, 1]
flag  = p_bad >= threshold
cost  = 5 * FN + FP
```

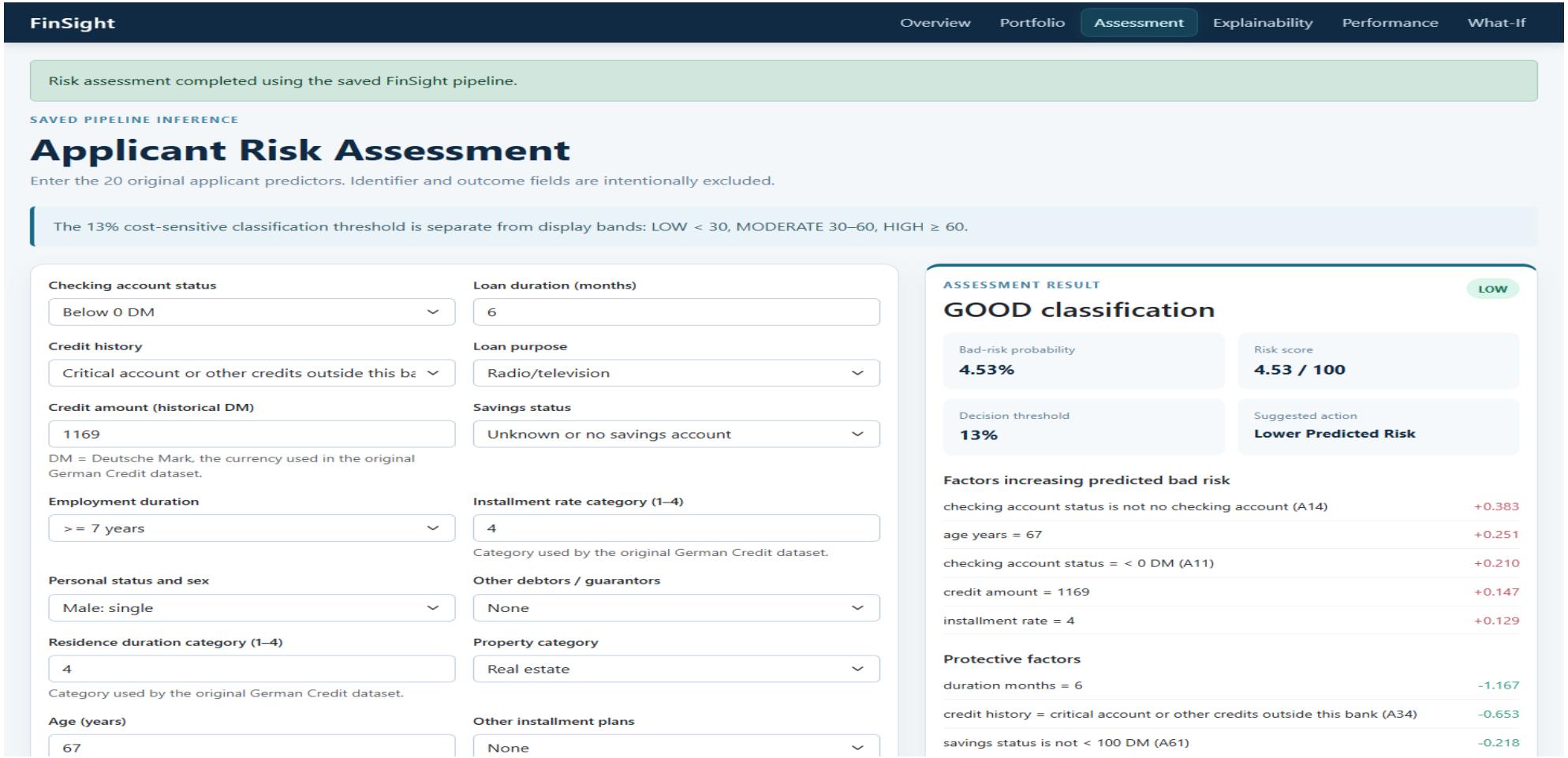
Working Prototype / Output Screens



Portfolio Analytics

Dataset and credit-risk portfolio overview

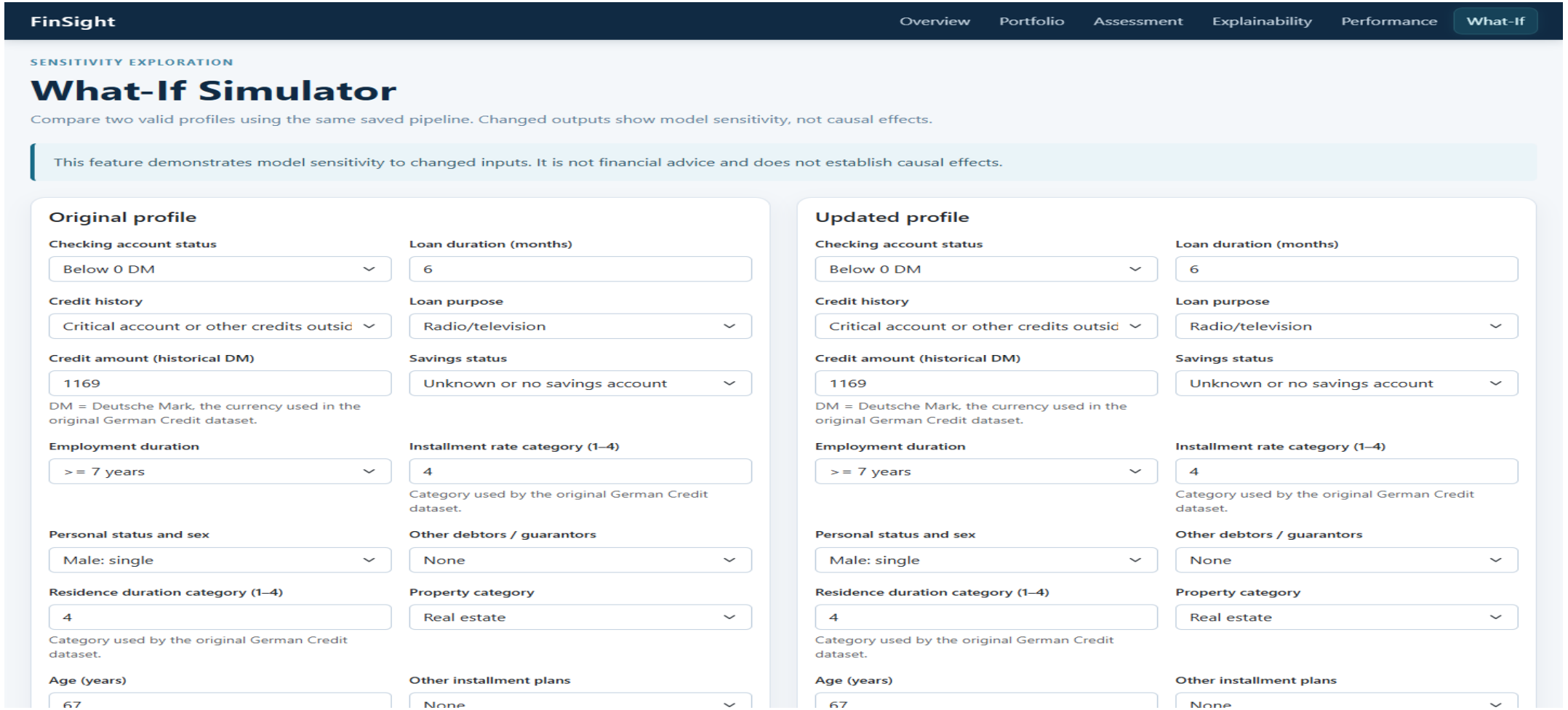
Working Prototype / Output Screens



Applicant Risk Prediction

Applicant-level probability, score and predicted risk

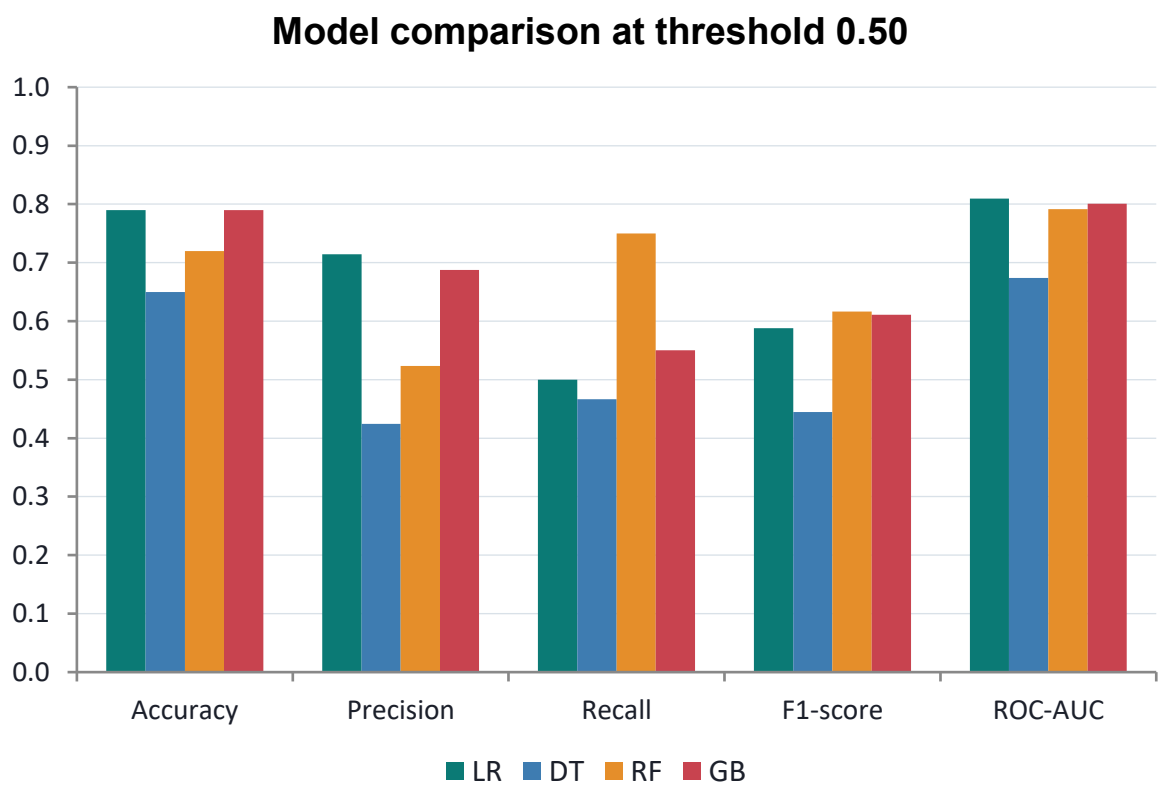
Working Prototype / Output Screens



Explainability / What-If Simulator

Feature explanation or sensitivity analysis after input changes

Results & Performance Comparison

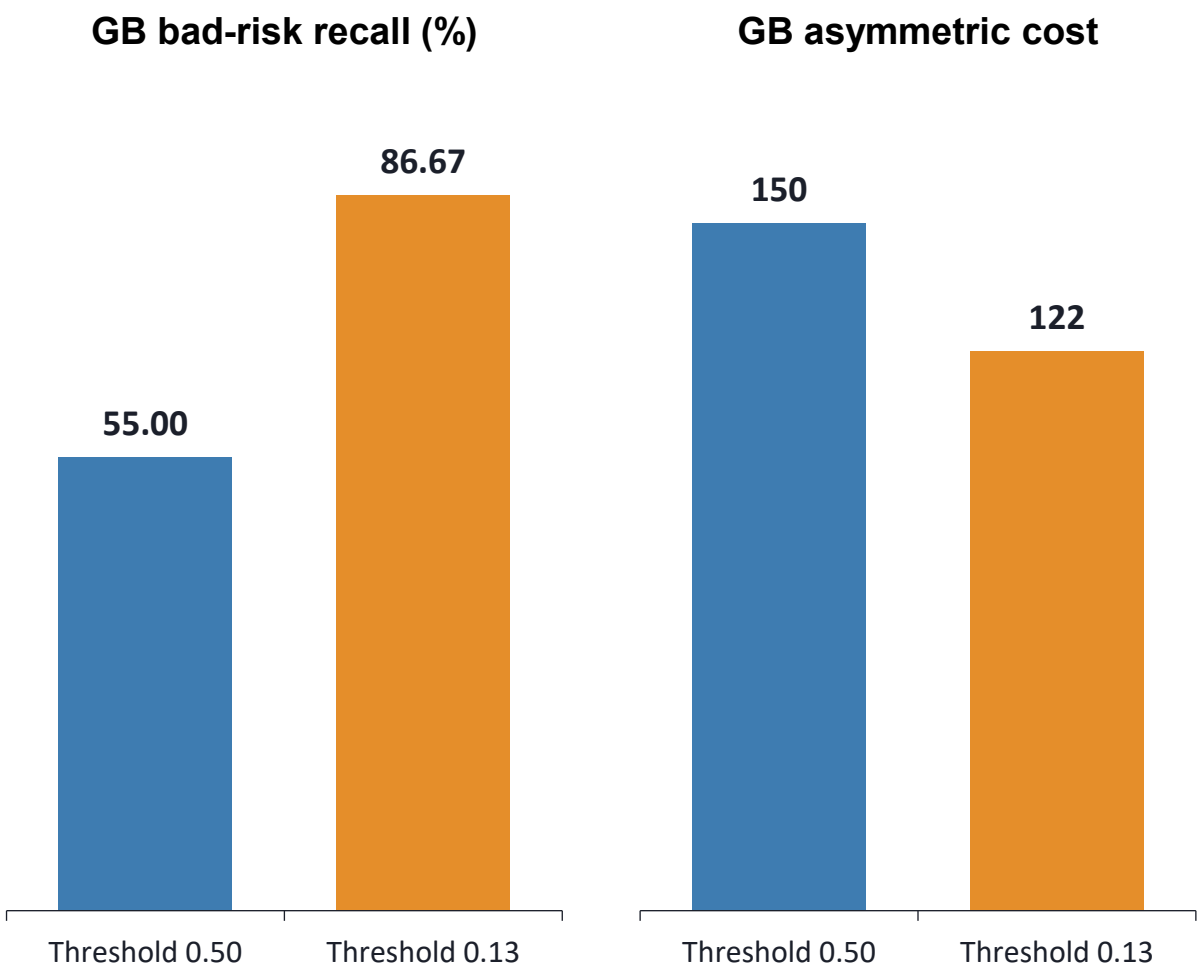


Metric	LR	DT	RF		
Accuracy	0.7900	0.6500	0.7200		
Precision	0.7143	0.4242	0.5233		
Recall	0.5000	0.4667	0.7500		
F1	0.5882	0.4444	0.6164		
ROC-AUC	0.8095	0.6741	0.7912		
Cost	162	198	116		

Selected from training-based cost analysis: Gradient Boosting

Thresholds: 0.50 for all models; GB @0.13 is the training-selected cost-aware threshold. Cost = 5·FN + FP. Logistic Regression had the lowest earlier holdout cost at its own training-selected threshold.

Analysis & Key Findings



86.67%

GB bad-risk recall at threshold 0.13 (up from 55.00%)

Headline: Cost-aware thresholding improves bad-risk detection but increases false positives.
Recall & cost: GB recall 55.00% → 86.67%; cost 150 → 122.
Trade-off: accuracy fell from 0.79 to 0.55.
Rankings: change with accuracy, recall, ROC-AUC or cost.
SHAP drivers: checking status, duration, purpose, credit history, savings, credit amount.

Testing & Validation

Test ID	Test Description	Expected Result	Actual Result	Status
TC-01	Dataset schema validation	20 predictors + valid target	Passed	Pass
TC-02	Bad-risk probability orientation	Class 1 = bad risk	Passed	Pass
TC-03	Risk-score boundary validation	Score within 0–100	Passed	Pass
TC-04	Flask prediction route	Valid inference response	Passed	Pass

Functional Testing

Pytest unit + integration testing

Validation Method

Stratified holdout and five-fold cross-validation

Test Suite Status

Earlier verified run: 81 passed, 0 failed

Challenges & Limitations

Challenges Faced

- Class imbalance → evaluated with recall, F1 and asymmetric cost
- Unequal error impact → introduced 5FN + FP cost function
- Data-leakage risk → preprocessing fitted only on training data

Limitations

- Small historical benchmark dataset
- Limited real-world generalization
- Probability calibration needs validation
- Fairness audit not yet complete
- SHAP is explanatory, not causal
- Production security needs further work

Conclusion

1. FinSight integrates preprocessing, four ML models, cost-aware evaluation, SHAP explainability and an interactive Flask decision-support system.

2. Cost-aware thresholding raised Gradient Boosting bad-risk recall from 55.00% to 86.67% and cut benchmark cost from 150 to 122, with more false positives.

3. Credit-risk evaluation should weigh asymmetric cost and explainability, not accuracy alone.



Final Outcome

86.67%

Bad-risk recall of the selected Gradient Boosting model at threshold 0.13.
Objectives achieved: 4 of 4

Future Scope & Applications

Improve Performance
Probability calibration, nested/repeated validation, broader threshold analysis

Expand Data
Corrected South German Credit, Taiwan credit-default, modern lending data

Deploy
Secure hosted service, API, enhanced reporting

Scale & Secure
Authentication, RBAC, fairness auditing, model monitoring, privacy controls

Applications

- Academic credit-risk analytics
- Decision-support experimentation
- Explainable ML research

Team Contribution

Reg. No.	Name	Contribution / Modules Handled
2420030475	M. Manasvini	Core ML pipeline, model training/comparison, cost-sensitive evaluation, threshold selection and final model integration
2420030267	N. Sreeja	Dataset validation, preprocessing, EDA and SHAP explainability
2420030548	S. Sreeja	Flask dashboard, applicant prediction, risk scoring, What-If Simulator and UI integration
2420030459	M. Sahasra	AWS S3/Glue/Athena integration, testing, validation and technical documentation

References

[1] N. Bussmann, P. Giudici, D. Marinelli, and J. Papenbrock, “Explainable Machine Learning in Credit Risk Management,” Computational Economics, vol. 57, pp. 203–216, 2021, doi: 10.1007/s10614-020-10042-0.

[2] S. M. Lundberg et al., “From local explanations to global understanding with explainable AI for trees,” Nature Machine Intelligence, vol. 2, pp. 56–67, 2020, doi: 10.1038/s42256-019-0138-9.

[3] X. Dastile and T. Celik, “Making Deep Learning-Based Predictions for Credit Scoring Explainable,” IEEE Access, vol. 9, pp. 50426–50440, 2021, doi: 10.1109/ACCESS.2021.3068854.

[4] A. Gramegna and P. Giudici, “SHAP and LIME: An Evaluation of Discriminative Power in Credit Risk,” Frontiers in Artificial Intelligence, vol. 4, Art. no. 752558, 2021, doi: 10.3389/frai.2021.752558.

[5] M. Bücker, G. Szepannek, A. Gosiewska, and P. Biecek, “Transparency, auditability, and explainability of machine learning models in credit scoring,” Journal of the Operational Research Society, vol. 73, no. 1, pp. 70–90, 2022, doi: 10.1080/01605682.2021.1922098.

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Project Links

GitHub:
https://github.com/Masetti07/DS_S7_T08_FinSight/tree/main
Dataset:
https://archive.ics.uci.edu/dataset/144/statlog+german+credit+data

THANK YOU

Questions & Discussion

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